p61 registered-root terminal closure

Reproducible run report · 23 October 2026

PASS **358,094,304 vertices accounted**

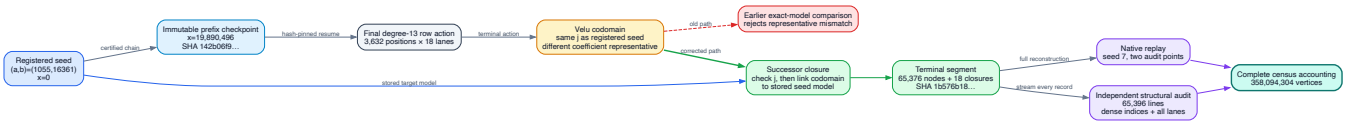
Curve `p61-i0025`
Field $p = 2^{61} - 1$
Model $y^2 = x^3 + 1055x + 16361$

Source `3b66e5cae94ab709…`
Run `run-20261023T200212Z`
Control plane v31, 89 pinned bindings

Outcome

The final row range $[19,890,496, 19,894,128)$ generated and replayed successfully. Across 18 lanes it contributes **65,376 vertices** and **18 closure certificates**. The immutable prefix plus this segment exactly matches the unconditionally certified class number of **358,094,304**.

Four layers support the result: successor generation, full native replay, digest-bound admission, and a separate Python 3.13 audit that streamed every record and checked dense coordinate/index occupancy, parent edges, and all closure lanes.



Exact accounting

Artifact	Bytes	SHA-256
Terminal segment	8,306,946	`1b576b189b7354bb9c025f17a14409cb…`
Checkpoint	7,311	`ef9c6857a84a497bc99a4de534f2cf54…`
Generation receipt	9,783	`eec5774ebd9ea7b0da959324bf3a6a52…`
Replay receipt	1,220	`94ec6052a4e6461f518bcee6637561c89…`
Admission manifest	1,943	`f3c03ef4965a4980e403d2e187ed4b45…`
Structural audit	2,521	`a165b95e5c26c9cad590a427fbc8b556…`

The segment expands to 36,207,183 bytes and contains 65,396 JSONL records: one header, 65,376 nodes, 18 closures, and one summary. The recomputed record chain is `c8a0499eea65009e87beba9b24d348b4…`.

Root cause and correction

The degree-13 action returned to the seed j-invariant. The earlier terminal check nevertheless rejected lane `(0,0)` because it compared a generic canonical coefficient pair with the separately registered root pair. The two models differ as coefficients but are isomorphic over F_p.

The successor selects and checks the target j-invariant, constructs the Elkies isogeny, and links the Velu codomain directly to the model stored at `x=0`. Checkpoint validation now permits a source transition only when the complete configured predecessor checkpoint digest matches.

Timing and controls

Stage	Elapsed	Control
Generation	35.134 s	8 configured threads; audit seed 0
Native replay	31.059 s	audit seed 7; two points
Structural follower	5.986 s	all 65,396 records

Dispatch occurred at one-minute load 10.6665 under the frozen limit 12.0, with 6.75 TB free under a 1.50 TB reserve. These are correctness-run timings, not a parallel-scaling benchmark.

Compatibility finding

The ordinary v1 follower correctly failed closed because it requires distinct producer/verifier binary digests and zero closure records. A terminal-specific v2 follower now binds its last passing ordinary receipt and checkpoint to the terminal structural audit. Managed control-plane v31 reports both the terminal generation gate and terminal audit follower as complete. The source-head serialized release-library suite passed 4,081 tests with zero failures and 114 ignored in 554.47 seconds. The rebased current-`main` code head then passed six focused tests and the complete serialized release-library suite: 4,479 passed, zero failed, 119 ignored in 592.14 seconds.

Evidence interpretation

Layer	What passed	What it establishes
Generation	65,376 nodes; 18 closures	The corrected terminal path constructs every lane
Native replay	Full segment reconstruction	Segment, checkpoint, kernels, links, and chain agree
Admission	All identities and digests bound	The predecessor-to-successor transition is immutable
Structural audit	Every JSONL record streamed	Dense occupancy, index map, parent edges, and closures agree

Scope boundary

This run resolves the registered-root census obstruction. It does not close global uniqueness, complete adjacency, endpoint resolution, reciprocal multiplicity, attack-label, held-out-split, null-control, or M7 confirmation gates. At the report snapshot, adjacency accounted for 37,199,232 of 2,506,660,128 directed edges. Every remaining censored or incomplete state must stay explicit in the training corpus.

Next transition

Gate	State	Required evidence
Census accounting	COMPLETE	358,094,304 admitted and terminal-audited
Global uniqueness	Active	External sort/index over every j-invariant
Selected adjacency	Active	2,506,660,128 directed edges
Endpoint and reciprocity	Waiting	Dense targets and exact multiplicities
M7 confirmation	Waiting	Masks, held-out splits, null controls, frozen evaluation